Iterating semi-discrete JKO steps by stochastic approximation

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(Euclidean) Gradient flows

$$E : \mathbb{R}^d \to \mathbb{R}, \qquad \dot{x}_t = -\nabla E(x_t)$$

- **Explicit Euler:** $\tau > 0$, $\nabla E(x_n^\tau) + \frac{x_{n+1}^\tau - x_n^\tau}{\tau} = 0$.
  Needs E to be differentiable.

- **Implicit Euler:** $\tau > 0$, $\nabla E(x_{n+1}^\tau) + \frac{x_{n+1}^\tau - x_n^\tau}{\tau} = 0$

- Equivalent to the variational problem

$$x_{n+1}^\tau = \operatorname*{arg\,min}_{x \in \mathbb{R}^d} E(x) + \frac{1}{2\tau}\|x - x_n^\tau\|^2$$

  Only needs a distance on $\mathbb{R}^d$!

(Wasserstein) Gradient flows

We define recursively, starting from $\mu \in \mathcal{P}_2(\mathbb{R}^d)$

$$\rho_0^\tau = \mu, \qquad \rho_{n+1}^\tau = \operatorname*{arg\,min}_{\rho \in \mathcal{P}_2(\mathbb{R}^d)} \mathcal{E}(\rho) + \frac{1}{2\tau} W_2(\rho, \rho_n^\tau)^2$$

- $W_2(\mu, \nu)$ is the optimal transport distance.

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- Piecewise constant interpolation of the $(\rho_n^\tau)_n$ and letting $\tau \to 0$ yields a curve in $\mathcal{P}_2(\mathbb{R}^d)$ solving (weakly) the continuity equation

  $$\partial_t \rho_t = -\mathrm{div}(\rho_t v_t)$$

  for some vector field v_t (related to $\mathcal{E}$).

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  for some vector field v_t (related to $\mathcal{E}$).
- $\mathcal{E}(\rho) = \int V\rho + \int \rho \ln(\rho)$ yields the Fokker–Planck equation

  $$\partial_t \rho_t = \Delta \rho_t + \mathrm{div}(x \rho_t)$$

One (general) JKO step 1/2

- Recall, given a time-step $\tau > 0$, we are interested in solving

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg\min_\rho \ \mathcal{H}(\rho|\gamma) + \dfrac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

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- **At step n:** We show that

$$\rho_{n+1}^\tau \propto \exp\left(-(\varphi^\star)^{c/2\tau}\right) \mathrm{d}\gamma$$

where $\varphi^\star$ is a Kantorovich potential satisfying

$$\varphi^\star \in \arg\min_\varphi \ \tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau.$$

One (general) JKO step 2/2

Proposition ***(G. 26+)***

The functional $\tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau$ is convex.

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Theorem *(G. 26+)*

Let $J(\varphi) := \int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma - \int \varphi \mathrm{d}\rho_n^\tau$. Then,

$$\arg\min_\varphi J(\varphi) \subsetneq \arg\min_\varphi \tilde{J}(\varphi) \quad \text{and} \quad \arg\min_{\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma = 1} \tilde{J}(\varphi) = \arg\min_\varphi J(\varphi).$$

Pro: J is "SGDisable"

Con: one more parameter to estimate (the normalisation constant!)

One semi-discrete JKO step 1/3

- $J(\varphi) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\varphi^{c/2\tau}(y)\right) - \int \varphi(x)\rho_n^\tau(\mathrm{d}x)\right]$

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$$\hat{\rho}_n = \frac{1}{N}\sum_{i=1}^{N}\delta_{x_i} \rightsquigarrow \begin{cases} \varphi(x_i) = \theta_i \\ \varphi^{c/2\tau}(y) = \min_i\left\{\frac{c(x_i,y)}{2\tau} - \theta_i\right\} \end{cases}$$

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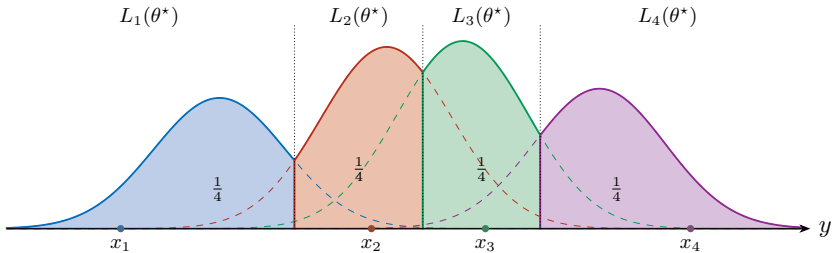
- $G(\theta) := J(\varphi(\cdot\,;\theta)) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(\max_{1\le i\le N}\left\{\theta_i - \frac{c(x_i,y)}{2\tau}\right\}\right) - \frac{1}{N}\sum_{i=1}^N \theta_i\right]$

One semi-discrete JKO step 2/3

- The next JKO step is

$$\rho^\star(y) \propto e^{\max_i \theta_i^\star - \frac{c(x_i,y)}{2\tau}} \gamma(y),$$

where $\theta^\star$ minimises G.

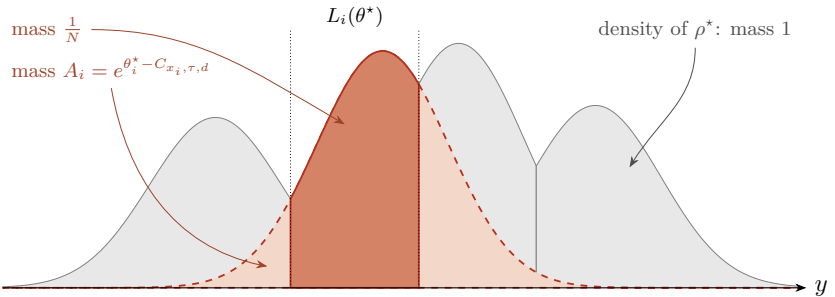


One semi-discrete JKO step 3/3

- The minimiser $\theta^\star$ is such that

$$\forall i, \qquad \rho^\star(L_i(\theta^\star)) = \frac{1}{N}.$$

- $\theta^\star \in \mathcal{B}$ which is explicit: $\forall i,\ \frac{1}{N} \leq A_i \leq 1$



Stochastic gradient descent 1/2

Let $q_s = \mathcal{N}(0, s^2 I_d)$ with $s^2 \geq \frac{\tau}{1+\tau}$ and $w_s(y) := \frac{\gamma(y)}{q_s(y)}$. Projected SGD algorithm:

$$\left\{ \begin{array}{l} \text{pick } y \sim q_s \text{ independent of } \theta^{(k)} \text{ and of } y_1, \dots, y_{k-1} \\ \text{let } i := \arg\min_j \ \{c_\tau(x_j, y_k) - \theta_j\} \\ \text{define } e_i \text{ as the } i\text{-th canonical basis vector of } \mathbb{R}^N \\ \theta^{(k+1)} := \mathrm{proj}_{\mathcal{B}} \left(\theta^{(k)} - \alpha_k w_s(y_k) e^{\theta_i^{(k)} - c_\tau(x_i, y_k)} e_i + \frac{\alpha_k}{N} \mathbb{1} \right) \end{array} \right. \tag{1}$$

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- There exists $B_s < +\infty$ such that $\mathbb{E}\|g_s(y;\theta)\|^2 \leq B_s^2$.

Stochastic gradient descent 2/2

Theorem *(G.)*

- $\alpha_k = \frac{2}{c_{\mathrm{qg}}(k+1)}, \quad \hat{\theta}_K := \frac{2}{K(K+1)} \sum_{k=1}^{K} k\theta^{(k)}$

Then, it holds that

$$0 \leq \mathbb{E}[G(\hat{\theta}_K)] - G(\theta^\star) \leq \frac{4B_s^2}{c_{\mathrm{qg}}(K+1)} = \mathcal{O}(K^{-1}),$$

where $\theta^\star$ denotes the unique minimiser of G.

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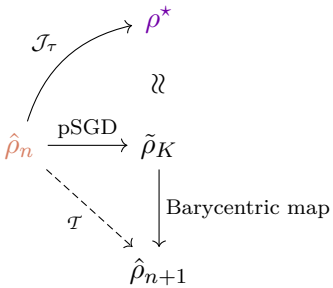
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Proposition *(G. 26+)*

Let $\mathcal{E} := \left\{ \|\nabla G(\hat{\theta}_K)\|_2 \leq \frac{1}{8} N^{-(1+\frac{1}{d})} \right\}$. On $\mathcal{E}$, every cell satisfies $\tilde{\rho}_K(L_i(\hat{\theta}_K)) \in \left[\frac{1}{2N}, \frac{3}{2N} \right]$, and

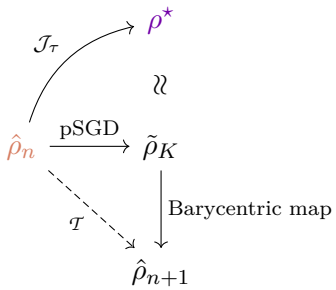
$$\mathbb{P}(\mathcal{E}^c) \leq \delta \qquad \text{as soon as} \qquad K \geq \frac{512 L_+ B_s^2 N^{2(1+\frac{1}{d})}}{c_{\text{qg}}\, \delta} - 1.$$

Back to a point cloud



- Recall that $\tilde{\rho}_K$ is absolutely continuous: to iterate, we need to discretise it back on N points.

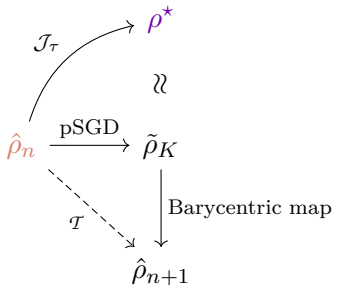
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- Recall that $\tilde{\rho}_K$ is absolutely continuous: to iterate, we need to discretise it back on N points.
- **One Lloyd step:** on $\mathcal{E}$ every cell has mass $\approx \frac{1}{N}$, so one representative per cell can carry mass $\approx \frac{1}{N}$. We choose the barycentres

$$\bar{x}_i := \frac{\int_{L_i(\hat{\theta}_K)} y \, \tilde{\rho}_K(\mathrm{d}y)}{\tilde{\rho}_K(L_i(\hat{\theta}_K))}, \qquad \hat{\rho}_{n+1} := \frac{1}{N} \sum_{i=1}^{N} \delta_{\bar{x}_i}.$$

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- **In practice:** importance sampling with the same q_s: $Y_1, \dots, Y_M \sim q_s$ and, for $Y_k \in L_i(\hat{\theta}_K)$, $w(Y_k) := w_s(Y_k)\, e^{\hat{\theta}_{K,i} - c_\tau(x_i, Y_k)}$,

$$\hat{x}_i = \frac{\sum_k Y_k \, w(Y_k) \mathbb{1}_{L_i(\hat{\theta}_K)}(Y_k)}{\sum_k w(Y_k) \mathbb{1}_{L_i(\hat{\theta}_K)}(Y_k)}, \qquad M \asymp N \ln(N)^{\frac{d+2}{2}}.$$

The barycentric map is bounded

Let $\mathfrak{B}$ be the barycentric map described in the previous slide: it sends the cloud $X = (x_1, \dots, x_N)$ to the cloud of barycentres of its Laguerre cells $L_i(\theta)$.

Theorem *(G. 26+)*

Let $R_\star := \frac{1+\tau}{\tau}\sigma\sqrt{2\ln(2N)}$ and let $R \geq R_\star$. If for all i, $L_i(\theta)$ has ρ_θ-mass $\geq \frac{1}{2N}$, then

$$X \in B(0,R)^N \implies \mathfrak{B}(X) \in B(0,R)^N.$$

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- Each barycentre is the mean of a Gaussian centred at $\frac{x_i}{1+\tau}$ conditioned on a set of mass at least $\frac{1}{2N}$. So, its norm is at most $\frac{R}{1+\tau} + \sigma\sqrt{2\ln(2N)}$, which is at most R exactly when $R \geq R_\star$.

Error of one approximate JKO step

Assume $\min_{i\neq j}\|x_i - x_j\| \geq CN^{-\frac{1}{d}}$ and let $\nu := \frac{1}{N}\sum_{i=1}^N \delta_{\bar{x}_i}$ (exact barycentres).

Theorem ***(Heuristic version, G. 26+)***

With $M \asymp N\ln(N)^{\frac{d+2}{2}}$ Monte Carlo samples,

$$\mathbb{E}[W_2(\hat{\rho}_{n+1}, \rho^\star)] \leq \underbrace{\mathbb{E}[W_2(\hat{\rho}_{n+1}, \nu)]}_{\text{Monte Carlo}} + \underbrace{\mathbb{E}[W_2(\nu, \tilde{\rho}_K)]}_{\text{quantisation}} + \underbrace{\mathbb{E}[W_2(\tilde{\rho}_K, \rho^\star)]}_{\text{pSGD}},$$

with Monte Carlo $\lesssim N^{-\frac{1}{2d}}\ln(N)^{\frac{d-1}{4}}$, quantisation $\lesssim N^{-\frac{1}{2d}}\ln(N)^{\frac{d+1}{4}}$ (both on $\mathcal{E}$) and pSGD $\lesssim C_N K^{-\frac{1}{4}}$, where C_N is polynomial in N.

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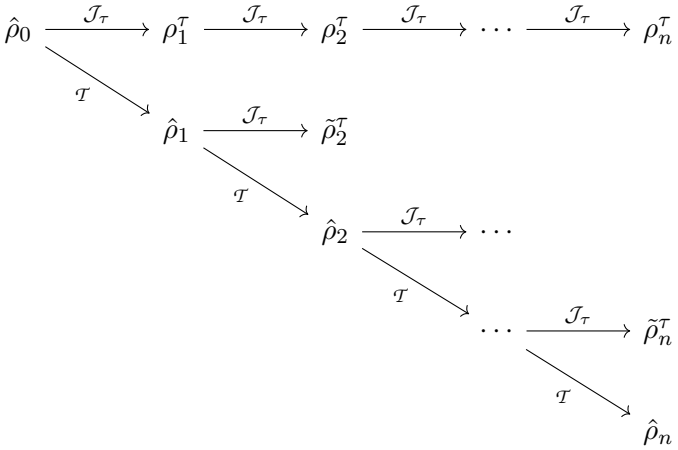
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with Monte Carlo $\lesssim N^{-\frac{1}{2d}}\ln(N)^{\frac{d-1}{4}}$, quantisation $\lesssim N^{-\frac{1}{2d}}\ln(N)^{\frac{d+1}{4}}$ (both on $\mathcal{E}$) and pSGD $\lesssim C_N K^{-\frac{1}{4}}$, where C_N is polynomial in N.

- Quantisation: extension of Mérigot–Santambrogio–Sarrazin (2021) to $\mathbb{R}^d$ and to cells of mass $\approx \frac{1}{N}$.
- With $K \gtrsim N^{3+\frac{7}{2d}}$ (up to logarithms), the one-step error is $\lesssim N^{-\frac{1}{2d}}\ln(N)^{\frac{d+1}{4}}$.

Iterating the whole procedure



Convergence to the Fokker–Planck flow

Theorem *(G. 26+)*

Let $(\varrho_t)_t$ solve Fokker–Planck with $\mathrm{KL}(\varrho_0|\gamma) < +\infty$. Under separation at every step and the budgets above, for all $n \geq 0$,

$$\mathbb{E}[W_2(\hat{\rho}_n, \varrho_{n\tau})] \leq \underbrace{(1+\tau)^{-\frac{n}{2}} W_2(\hat{\rho}_0, \varrho_0)}_{\text{initial error}} + \underbrace{\frac{1+\tau+\sqrt{1+\tau}}{\tau}\,\bar{\eta}}_{\text{space}} + \underbrace{c_{n,\tau}\sqrt{\tau\,\mathrm{KL}(\varrho_0|\gamma)}}_{\text{time}},$$

$\bar{\eta}$: bound on the one-step error; $c_{n,\tau}$: explicit, bounded uniformly in n for $\tau \leq 1$.

- No accumulation: $\frac{1+\tau+\sqrt{1+\tau}}{\tau} = \sum_{j\geq 0}(1+\tau)^{-\frac{j}{2}}$, and the space term is $\lesssim \tau^{-1-\frac{d+1}{4}} N^{-\frac{1}{2d}}$ up to logarithms.

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- Vanishes uniformly in n as $W_2(\hat{\rho}_0, \varrho_0) \to 0$, $N \to \infty$, $\tau \to 0$ with $\tau \geq N^{-\frac{2}{d(d+5)}+\epsilon}$.

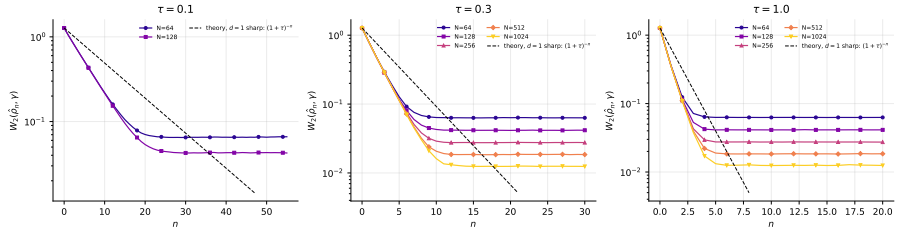
In practice

The implemented solver departs from the analysed one:

- fixed maximal budget $K = 400{,}000$, with an early stop once every cell has mass $\geq \frac{3}{4N}$;
- an extra mass-adaptation step (cyclic coordinate descent);
- barycentres estimated from samples of γ;
- warm start at the previous potential, suffix averaging of the iterates.

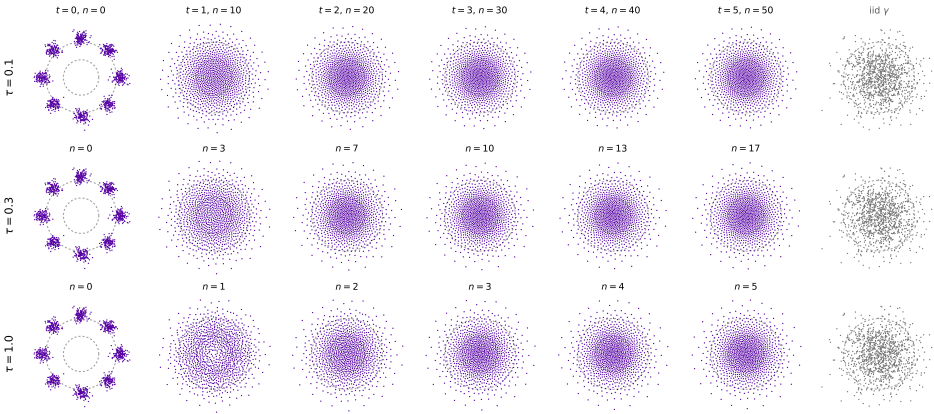
The theorems do not cover this solver: the experiments compare its behaviour with the predicted rates.

Numerical experiments: $d = 1$



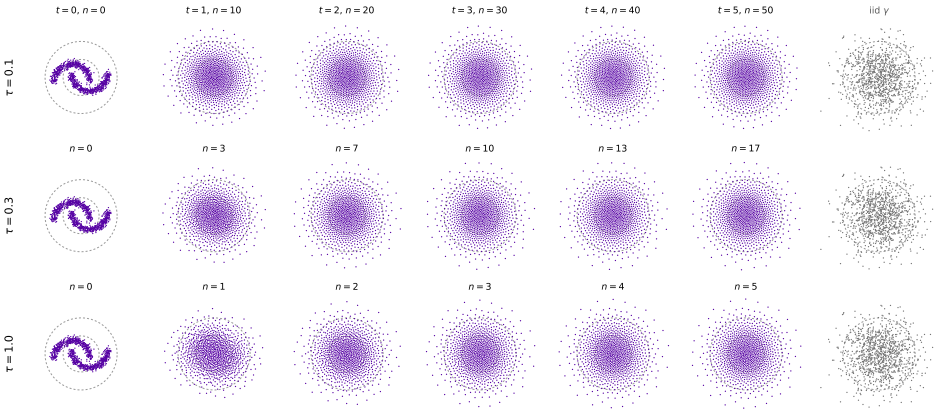
Initial measure: balanced mixture of two Gaussians (centres ± 2, standard deviation 0.2). Dashed: the contraction rate $(1+\tau)^{-n}$ of the JKO scheme in dimension one.

Numerical experiments: $d = 2,\ N = 1024$



One row per $\tau \in \{0.1, 0.3, 1\}$, one column per physical time $t = n\tau$. Last column: an i.i.d. draw from γ of the same size.

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